

Kharmen Calibration Builder User Manual

Setup, calibration review, masking, KDE review, and pipeline handoff

Prepared for internal AutoSpect / LA-ICP-TOF-MS calibration workflows
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Purpose

This manual explains how to use the Kharmen Calibration Builder Setup GUI and the Calibration Review GUI to connect ICP/reference standards to AutoSpect-fitted signals, assign files and standards, apply standard-only correction options, inspect masks and outliers, and push reviewed data to KDE and the Species Surface Editor.

Contents

1. Quick start

2. Workflow overview

3. Setup GUI reference

3.1 ICP / reference table tab

3.2 Files tab

3.3 Processing / correction tab

3.4 Run / profile tab

4. Calibration Review GUI reference

4.1 Outlier Inspector tab

4.2 KDE tab

4.3 Species Surface Editor

5. Basic workflows

6. Practical notes and troubleshooting

7. Checklists

1. Quick start

Use this section when you already have AutoSpect-processed calibration files and an ICP/reference table ready.

Before you begin: required file organization

Organize the AutoSpect-fitted calibration files into separate folders before launching the calibration builder. The main calibration standards should be in their own folder. The standards measured before the sample should be in a separate before-run folder, and the standards measured after the sample should be in a separate after-run folder. Keeping these folders separate allows the GUI to assign calibration roles correctly when the files are scanned.

A note on file nomenclature (optional): File nomenclature can help in quick assortment: the current version filters the file names as follows: LA[any characters except a space] [concentration] [trial]_[any other parts of the name]. For example, in the file name “LA_Na 5000 6_2025-02-04_16h40m20s” the expected calibrant concentration is 5000 and the trial number is 6.

1. Open the setup GUI from MATLAB with `asv_MassCal4Kharmen_gui`.
2. On the ICP / reference table tab, browse to the Excel reference table and press Import.
3. On the Files tab, choose the main calibration, before-run standard, and after-run standard folders, then press Scan files.
4. Review the file table. Correct the Group and Standard ID assignments using the bulk controls when needed.
5. On the Processing / correction tab, select the calibration elements, confirm the AutoSpect species names, and choose S or P as the standard-only correction element.
6. On the Run / profile tab, choose whether to rebuild or reuse the loaded-input cache, then press Run Kharmen analysis.
7. In the Calibration Review window, inspect the Outlier Inspector tab, adjust masks and filtering, and press “Process to Pipeline” when the current review state is ready.
8. Inspect the KDE tab and Species Surface Editor. Save or reload sessions from the Kharmen Session menu or Load/refine calibration session workflow.

Recommended first run

For a new dataset, enable Rebuild cache on this run. Use the loaded-input cache only after confirming the same files and assignments are being reused.

2. Workflow overview

The GUI separates setup from review. The setup window defines standards, files, correction settings, and cache behavior. The review window is used to inspect outliers, control masks, review KDE distributions, and push the reviewed data to the calibration surface workflow.

Stage	Main action	Output
Reference table	Import the ICP/reference Excel table.	Standard IDs and element reference values.
File assignment	Scan AutoSpect files and assign Group / Standard ID / Trial.	A file table linking each .mat file to a standard role.
Processing setup	Choose elements, AutoSpect species, S/P correction, and display options.	A configured calibration run.
Run/profile	Choose cache/profile settings and launch the analysis.	Loaded inputs and a Calibration Review session.
Outlier inspection	Inspect maps, histograms, traces, and calibration summaries.	Reviewed masks and calibrant inclusion/exclusion choices.
Pipeline handoff	Press Process to Pipeline.	Updated KDE view and Species Surface Editor input.

3. Setup GUI reference

The Kharmen Calibration Builder Setup window contains four tabs. Complete the tabs from left to right for a new run.

3.1 ICP / reference table tab

This tab defines the known reference values used for calibration. The first column is the Standard ID or standard name; columns 2:end contain element-specific ICP/reference values.

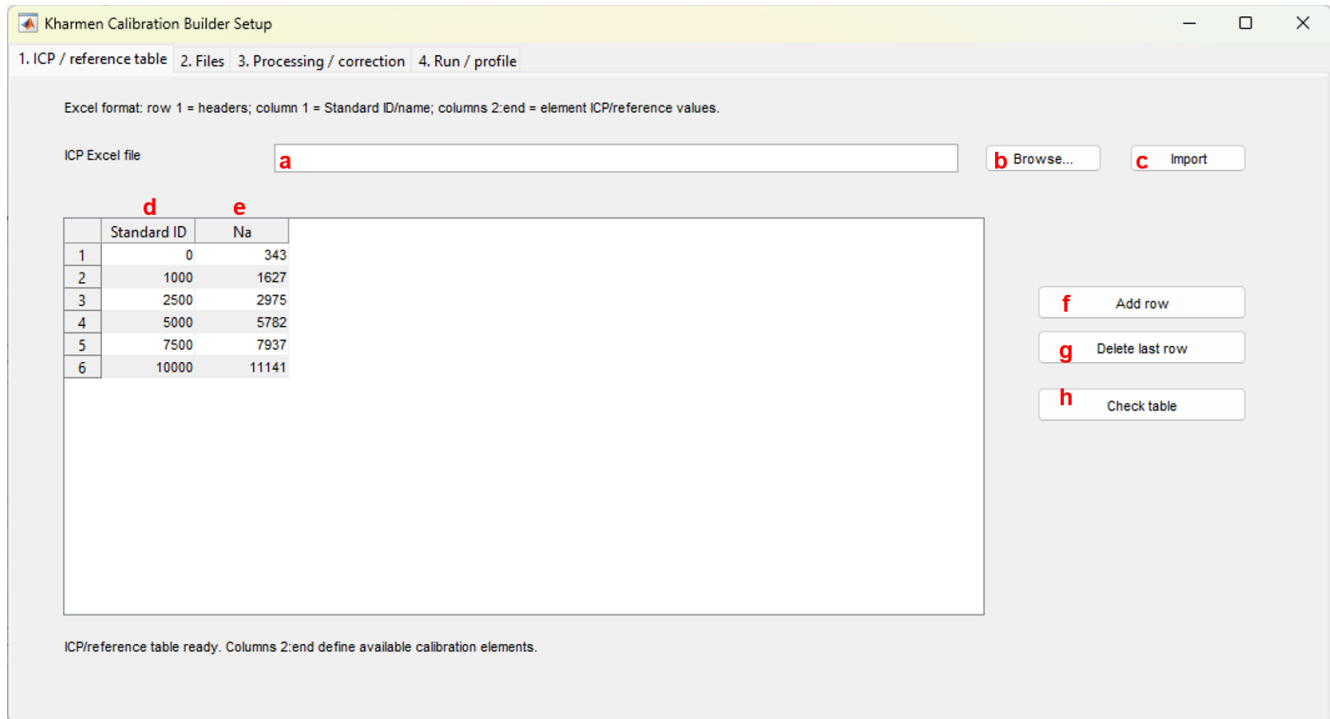


Figure 1. ICP / reference table tab with annotated controls.

Label	Control	Description
a	ICP Excel file path	Displays the selected Excel reference file.
b	Browse...	Opens a file picker to select the ICP/reference Excel file.
c	Import	Imports the selected reference table into the GUI.
d	Standard ID column	Defines the standard levels or names used to match calibration files.
e	Element/reference columns	Reference values for each calibration element.
f	Add row	Adds a blank row for manual editing.
g	Delete last row	Deletes the final row in the reference table.
h	Check table	Validates the current reference table format and values.

Required Excel format

Row 1 must contain headers. Column 1 must contain Standard ID/name. Columns 2:end must contain ICP/reference values. The Standard IDs used here must match the file assignments on the Files tab.

Example spread sheet

	Na	Mg	P	S	K	Ca	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	As	Se
IV4_12.5		11.767		336.719	13.223	13.369	11.633	11.673	11.33	11.905	11.963	11.638	12.185	11.834	12.742	13.926
IV4_25		25.49		356.775	26.254	26.721	24.832	25.901	24.608	25.762	26.133	25.346	26.449	25.736	27.815	29.291
IV4_50		48.012		372.938	49.245	49.558	49.308	48.589	48.353	48.521	48.699	50.235	50.709	48.521	52.302	54.947
blank	34.003	0.098	26.565	340.402	0.652	1.341	0	0.005	0	0.058	0	0	0.007	0.048	0.002	0.025
NaP_1000	706.665		948.946													
NaP_2000	1443.09		1995.94													
NaP_4000	2846.53		3999.84													

Row 1 must contain headers. Column 1 must contain the Standard ID/name. Columns 2+ must contain ICP/reference values for each element. Standard IDs used in this spreadsheet must match the file assignments in the GUI. Blank

cells should be left blank when a calibrant does not contain a given element; these blanks are treated as unavailable calibration values, not as zero concentration.

3.2 Files tab

The Files tab locates AutoSpect output files and assigns each file to a calibration role. The GUI can infer many assignments from filenames, but the table should always be reviewed before running the analysis.

Hint: The three folder selectors are intended for three distinct file groups: main calibration standards, before-run standards, and after-run standards. Do not mix these files in one folder unless you plan to manually correct the assignments after scanning.

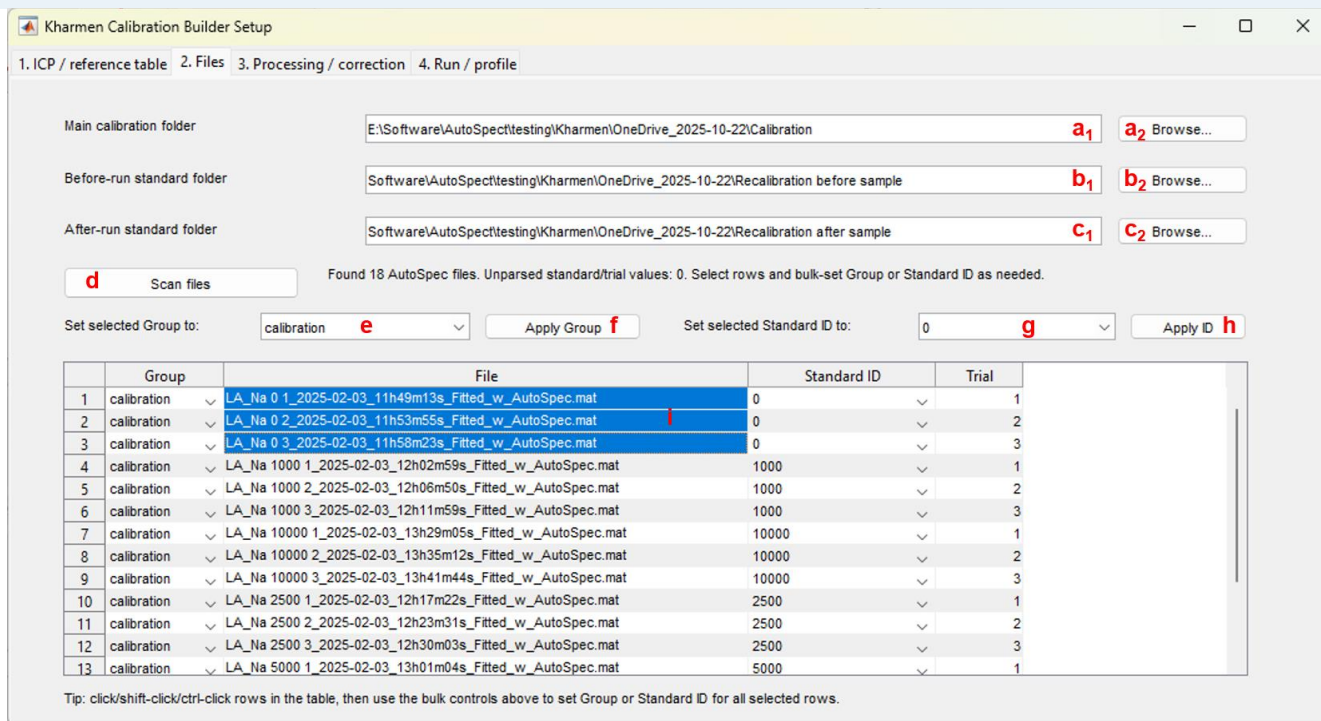


Figure 2. Files tab with folder selection, scan controls, and file-assignment table.

Label	Control	Description
a1	Main calibration folder	Folder containing the primary calibration standards.
a2	Browse...	Selects the main calibration folder.
b1	Before-run standard folder	Folder containing standards measured before sample acquisition.
b2	Browse...	Selects the before-run folder.
c1	After-run standard folder	Folder containing standards measured after sample acquisition.
c2	Browse...	Selects the after-run folder.
d	Scan files	Searches the selected folders for AutoSpect .mat files.
e	Set selected Group to	Chooses a group for the selected table rows.
f	Apply Group	Applies the selected group to highlighted rows.
g	Set selected Standard ID to	Chooses a Standard ID for highlighted rows.
h	Apply ID	Applies the selected Standard ID to highlighted rows.

Table column	Purpose
Group	Calibration role, typically calibration, before, or after.
File	AutoSpect .mat file name.
Standard ID	Reference-table standard assigned to the file.
Trial	Replicate or trial number inferred from the file name or file order.

3.3 Processing / correction tab

This tab controls which elements are calibrated, which AutoSpect species is used for each element, and how standard-only correction is handled.

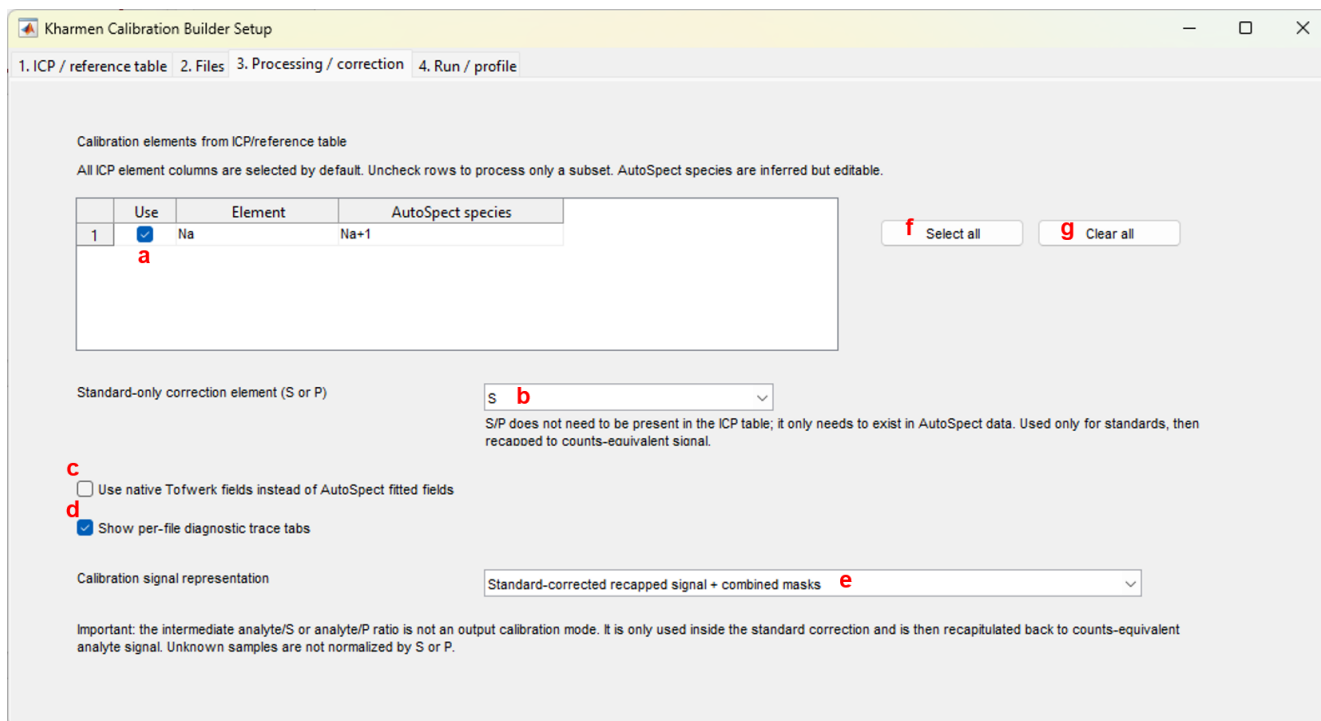


Figure 3. Processing / correction tab with element selection and correction controls.

Label	Control	Description
a	Use checkbox	Enables or disables an element for calibration processing.
b	Standard-only correction element	Selects S or P as the correction element for standards.
c	Use native Tofwerk fields instead of AutoSpect fitted fields	Uses native fields instead of AutoSpect fitted fields. Leave unchecked unless specifically required.
d	Show per-file diagnostic trace tabs	Displays detailed diagnostic trace tabs for individual files.
e	Calibration signal representation	Chooses the signal representation used for calibration. The default standard-corrected recapped signal remains counts-equivalent.
f	Select all	Enables all element rows.
g	Clear all	Disables all element rows.

Important: S/P correction is standard-only

The intermediate analyte/S or analyte/P ratio is not an output calibration mode. It is used inside the standard correction and then recapped back to a counts-equivalent analyte signal. Unknown samples are not normalized by S or P.

3.4 Run / profile tab

The Run / profile tab controls cache behavior, reusable setup profiles, and loading/refining saved calibration sessions.

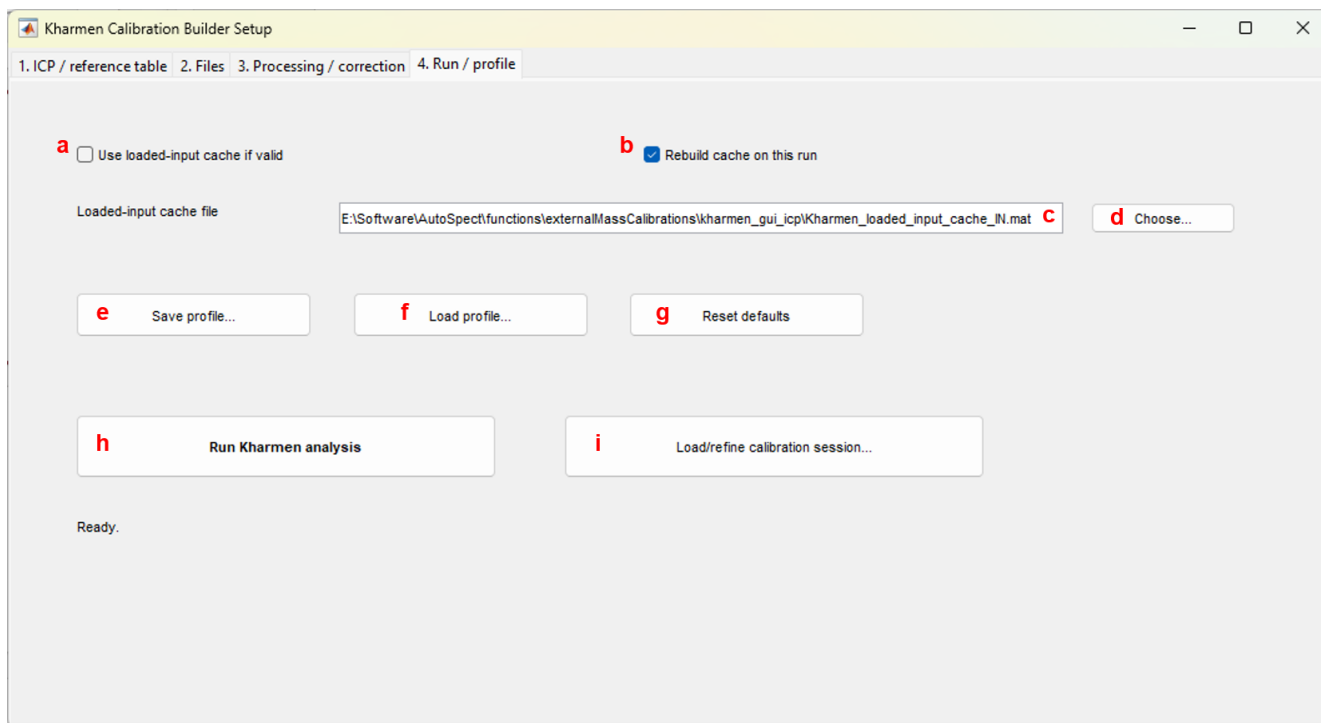


Figure 4. Run / profile tab with corrected control labels.

Label	Control	Description
a	Use loaded-input cache if valid	Reuses an existing loaded-input cache when it matches the current inputs.
b	Rebuild cache on this run	Forces the GUI to reload and rebuild cached input data. Recommended for a first run or when files changed.
c	Loaded-input cache file	Displays the cache file path.
d	Choose...	Selects the loaded-input cache file location.
e	Save profile...	Saves the current setup options.
f	Load profile...	Loads a previously saved setup profile.
g	Reset defaults	Restores default GUI settings.
h	Run Kharmen analysis	Starts the calibration-processing workflow and opens the review window.
i	Load/refine calibration session...	Loads a saved review session for continued refinement.

4. Calibration Review GUI reference

The Calibration Review window opens after the setup analysis is launched. It contains the Outlier Inspector, KDE, and Species Surface Editor tabs. The Activity log reports major processing and review actions.

4.1 Outlier Inspector tab

Use the Outlier Inspector to inspect raw/corrected calibration signals, define masks, inspect trace behavior, and decide when to push the current reviewed state downstream.

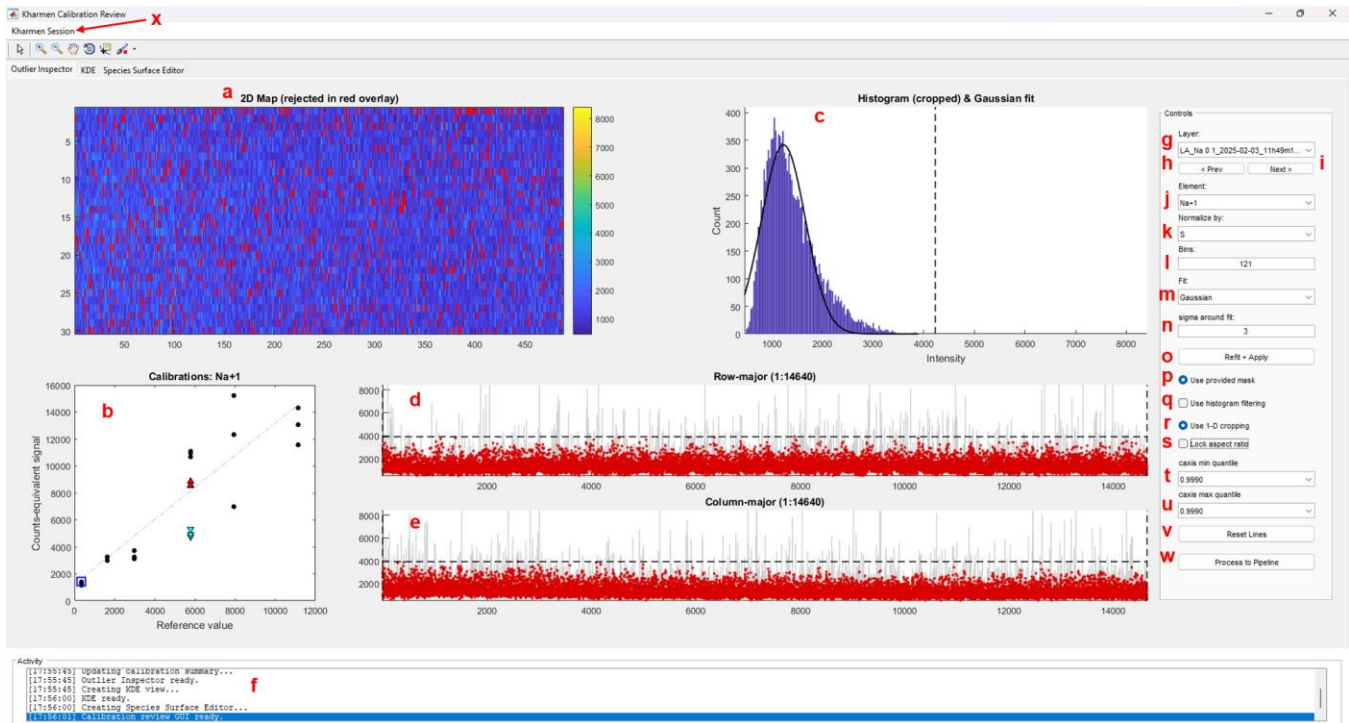


Figure 5. Outlier Inspector tab with image, histogram, traces, controls, and Activity log.

Label	Panel/control	Description
a	2D Map	Displays the selected file/layer as an image. Rejected pixels are overlaid in red.
b	Calibration summary	Shows calibration points against reference values for the selected element.
c	Histogram and fit	Displays the signal histogram, fit model, and crop/threshold lines.
d	Row-major trace	Shows signal values in row-major order.
e	Column-major trace	Shows signal values in column-major order.
f	Activity log	Displays major GUI actions and processing status.
g	Layer	Selects the current file/layer.
h/i	Prev / Next	Moves to the previous or next layer.
j	Element	Selects the calibration element/species.
k	Normalize by	Chooses Nothing, S, or P for review display.
l	Bins	Controls the number of histogram bins.
m	Fit	Selects the histogram fit model.
n	Sigma around fit	Controls the accepted range around the fitted distribution.
o	Refit + Apply	Refits the histogram and applies the current filter settings.
p	Use provided mask	Uses the mask supplied by the input data.
q	Use histogram filtering	Enables histogram-based rejection.
r	Use 1-D cropping	Enables one-dimensional signal cropping.
s	Lock aspect ratio	Keeps the map aspect ratio locked.
t/u	caxis min/max quantile	Controls image display contrast using lower and upper quantiles.
v	Reset Lines	Resets crop/threshold display lines.
w	Process to Pipeline	Pushes the current reviewed state to KDE and Species Surface Editor.
x	Menu	Save the session so it can be loaded as an external calibration in InSpect and/or so it can be reopened later for further optimization.

Responsiveness behavior

Mask and filter edits are reviewed in the Outlier Inspector first. KDE and Species Surface Editor updates are deferred until Process to Pipeline is pressed, so the GUI remains responsive during repeated filtering adjustments.

4.2 KDE tab

The KDE tab displays standard-level signal distributions after the current reviewed state has been sent through the pipeline. The tab supports both raw-only 1-D plots and S/P-based 2-D KDE plots.

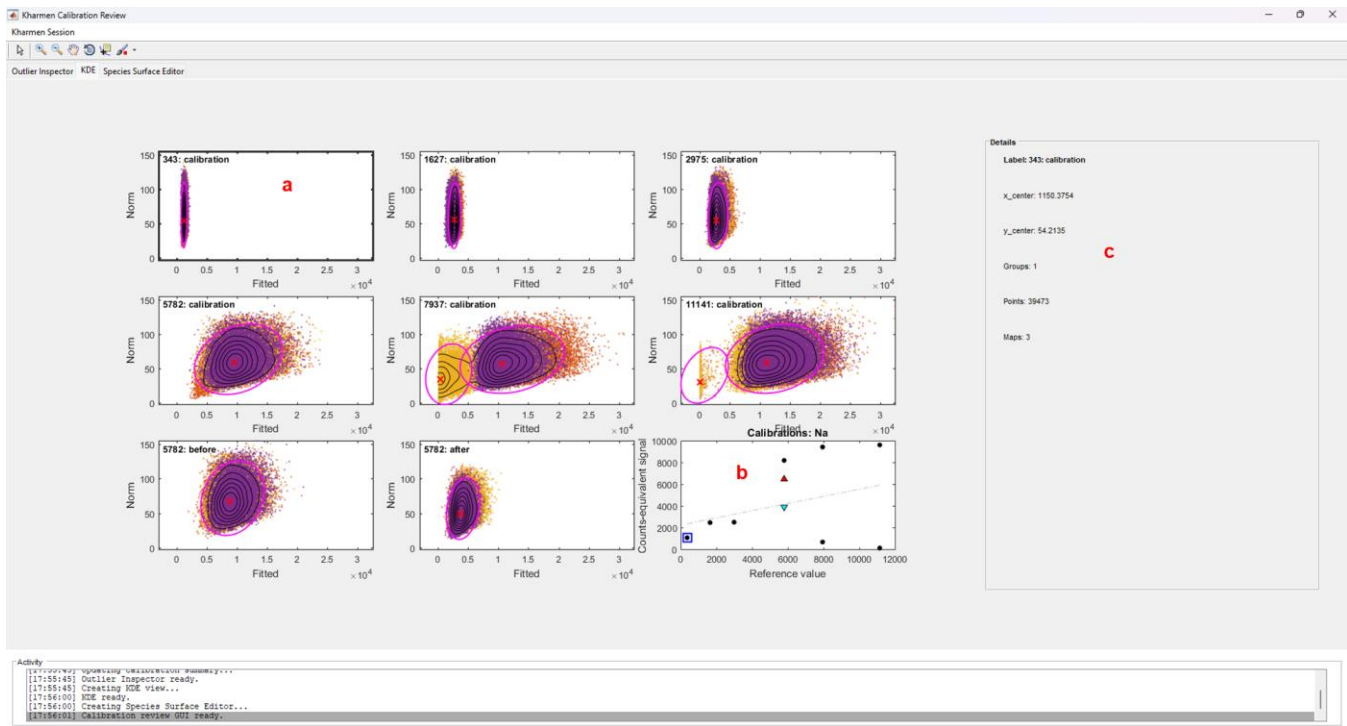


Figure 6. KDE tab showing 2-D KDE distributions, calibration summary, and details panel.

Label	Panel	Description
a	KDE / distribution plots	Shows per-standard or per-group distributions. When Normalize by is S or P, the 2-D raw-vs-normalization KDE layer is visible. When Normalize by is Nothing, the 1-D raw-only layer is visible.
b	Calibration summary	Shows the current calibration relationship after reviewed masks and exclusions are applied.
c	Details panel	Displays metadata for the selected distribution or calibration point.

KDE display modes

The KDE tab keeps both display systems available. Normalize by Nothing shows the 1-D raw-only plots. Normalize by S or P shows the 2-D raw-vs-normalization KDE plots. Switching modes toggles visibility rather than rebuilding the full tab.

4.3 Species Surface Editor

The Species Surface Editor receives the processed calibration data after Process to Pipeline is pressed. Use it to inspect and refine the calibration surface for the selected element/species. The review GUI pushes the current masks, calibrant exclusions, and signal representation downstream only when the current review state is processed to the pipeline.

Action	Effect
Work in Outlier Inspector	Edits masks, filters, and exclusions without forcing full downstream recalculation after each small change.
Press Process to Pipeline	Updates KDE and pushes the reviewed data to the Species Surface Editor.
Save calibration session	Preserves the review state for later refinement.

5. Basic workflows

5.1 Build a new calibration from scratch

1. Start asv_MassCal4Kharmen_gui.
2. Import the ICP/reference table.
3. Scan the calibration, before-run, and after-run folders.
4. Correct Group and Standard ID assignments.
5. Select calibration elements and confirm AutoSpect species names.
6. Choose the standard-only correction element, typically S or P.
7. Choose cache behavior and run the analysis.
8. Inspect the Outlier Inspector tab and refine masks or exclusions.
9. Press Process to Pipeline.
10. Review KDE and the Species Surface Editor, then save the session if needed.

5.2 Correct Standard ID assignments

1. Go to the Files tab.
2. Select one or more rows in the file table.
3. Choose the desired Standard ID from Set selected Standard ID to.
4. Press Apply ID.
5. Repeat until all files match the Standard IDs in the reference table.

5.3 Change file group assignments

1. Go to the Files tab.
2. Select one or more file rows.
3. Choose calibration, before, or after from Set selected Group to.
4. Press Apply Group.

5.4 Process only a subset of elements

1. Go to Processing / correction.
2. Uncheck elements that should not be processed.
3. Use Select all or Clear all for bulk changes.
4. Confirm the AutoSpect species names before running the analysis.

5.5 Review masks without slowing down KDE

1. Stay in the Outlier Inspector tab while adjusting masks, histogram filtering, cropping, or fit settings.
2. Use the map, histogram, row-major trace, column-major trace, and calibration summary to judge the current state.
3. Press Process to Pipeline only when the current masks and settings should update KDE and the Species Surface Editor.

5.6 Switch between raw and S/P review

1. Use the Normalize by dropdown in the Outlier Inspector.
2. Choose Nothing for raw-only review.
3. Choose S or P to inspect standard-only correction behavior.
4. Press Process to Pipeline when the selected state should be pushed downstream.

6. Practical notes and troubleshooting

Issue or question	Recommendation
Standard IDs do not match	Edit the Files table so each assigned Standard ID matches a Standard ID in the ICP/reference table.
A file is assigned to the wrong group	Select the row, choose the correct group, and press Apply Group.
Stale data appear after changing folders or files	Enable Rebuild cache on this run and rerun the analysis.
KDE or Species Surface Editor does not reflect recent mask edits	Press Process to Pipeline from the Outlier Inspector.
The GUI is slow during review	Avoid pushing to the pipeline after every small edit. Refine masks first, then press Process to Pipeline once.
Need to reuse settings later	Use Save profile for setup options and save a calibration session for reviewed masks/exclusions.

Cache guidance

Rebuild cache when files, folders, assignments, or AutoSpect outputs changed. Reuse a valid loaded-input cache when the same files and setup are being rerun.

7. Checklists

7.1 Before running analysis

- Main calibration files are in their own folder.
- Before-run standard files are in their own folder.
- After-run standard files are in their own folder.
- ICP/reference table imported correctly.
- Standard IDs are correct and match file assignments.
- Element reference columns are correct.
- Calibration files were scanned successfully.
- Groups are assigned correctly.
- Standard IDs are assigned correctly.
- Calibration elements are selected.
- AutoSpect species names are correct.
- Standard-only correction element is set appropriately.
- Cache behavior is appropriate for the run.

7.2 Before saving a reviewed calibration session

- Outlier masks look reasonable.
- Histogram/cropping thresholds are acceptable.
- Calibration summary points are reasonable.
- Bad calibrants are excluded where appropriate.
- Process to Pipeline has been pressed after the final review edits.
- KDE view has been reviewed.
- Species Surface Editor has been reviewed.